```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
df = sns.load_dataset('mpg').dropna()
df.head()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino

Next steps: Generate code with df New interactive sheet

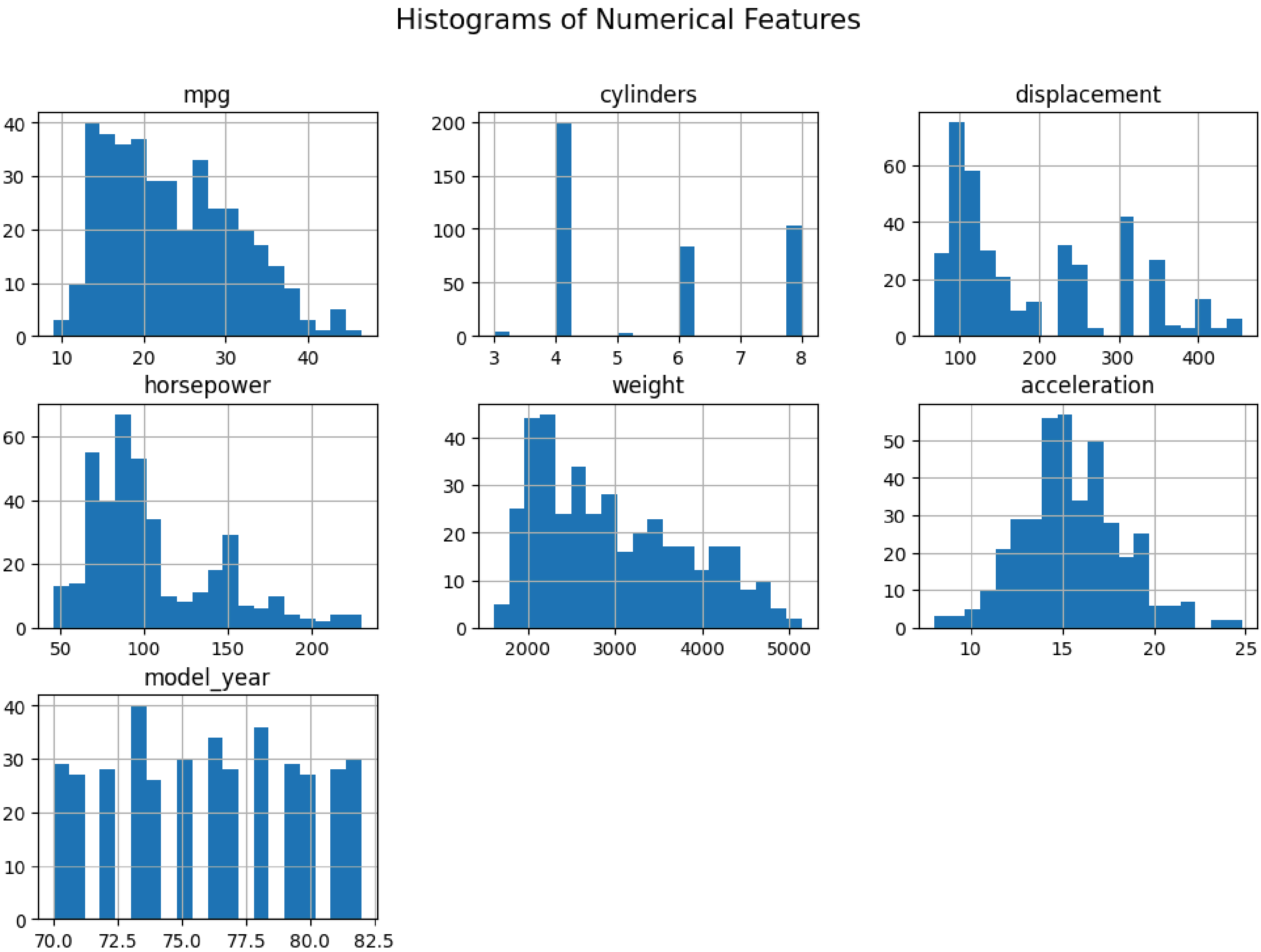
```
print("Dataset Info:")
print(df.info())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
Index: 392 entries, 0 to 397
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   mpg           392 non-null    float64
 1   cylinders     392 non-null    int64  
 2   displacement  392 non-null    float64
 3   horsepower    392 non-null    float64
 4   weight        392 non-null    int64  
 5   acceleration  392 non-null    float64
 6   model_year    392 non-null    int64  
 7   origin        392 non-null    object 
 8   name          392 non-null    object 
dtypes: float64(4), int64(3), object(2)
memory usage: 30.6+ KB
None
```

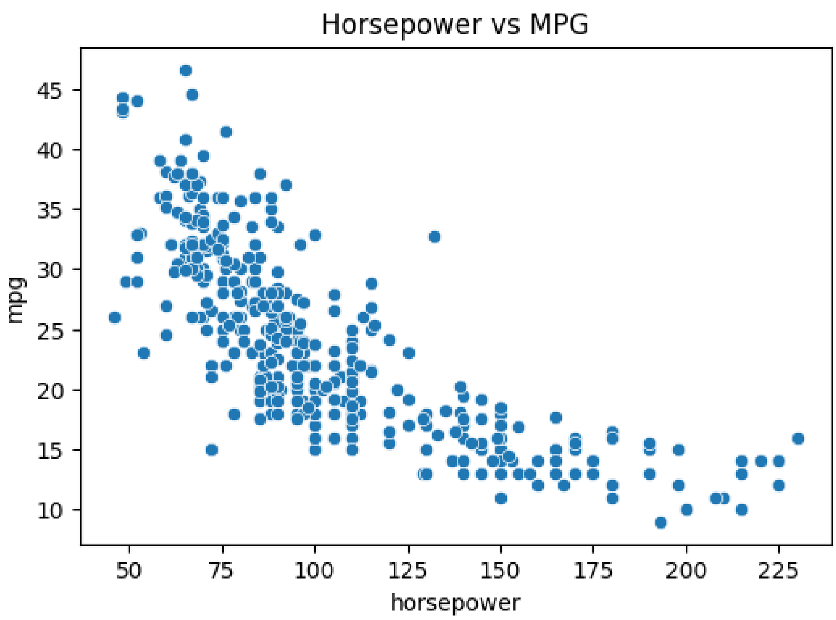
```
print("\nSummary Statistics:")
print(df.describe())
```

```

Summary Statistics:
              mpg   cylinders  displacement  horsepower       weight  \
count  392.000000  392.000000    392.000000  392.000000   392.000000   
mean    23.445918    5.471939    194.411990  104.469388  2977.584184   
std      7.805007    1.705783    104.644004   38.491160   849.402560   
min      9.000000    3.000000     68.000000   46.000000  1613.000000   
25%     17.000000    4.000000    105.000000   75.000000  2225.250000   
50%     22.750000    4.000000    151.000000   93.500000  2803.500000   
75%     29.000000    8.000000    275.750000  126.000000  3614.750000   
max     46.600000    8.000000    455.000000  230.000000  5140.000000   

       acceleration  model_year  
count    392.000000  392.000000  
mean      15.541327   75.979592  
std        2.758864    3.683737  
min        8.000000   70.000000  
25%       13.775000   73.000000  
50%       15.500000   76.000000  
75%       17.025000   79.000000  
max       24.800000   82.000000  
```

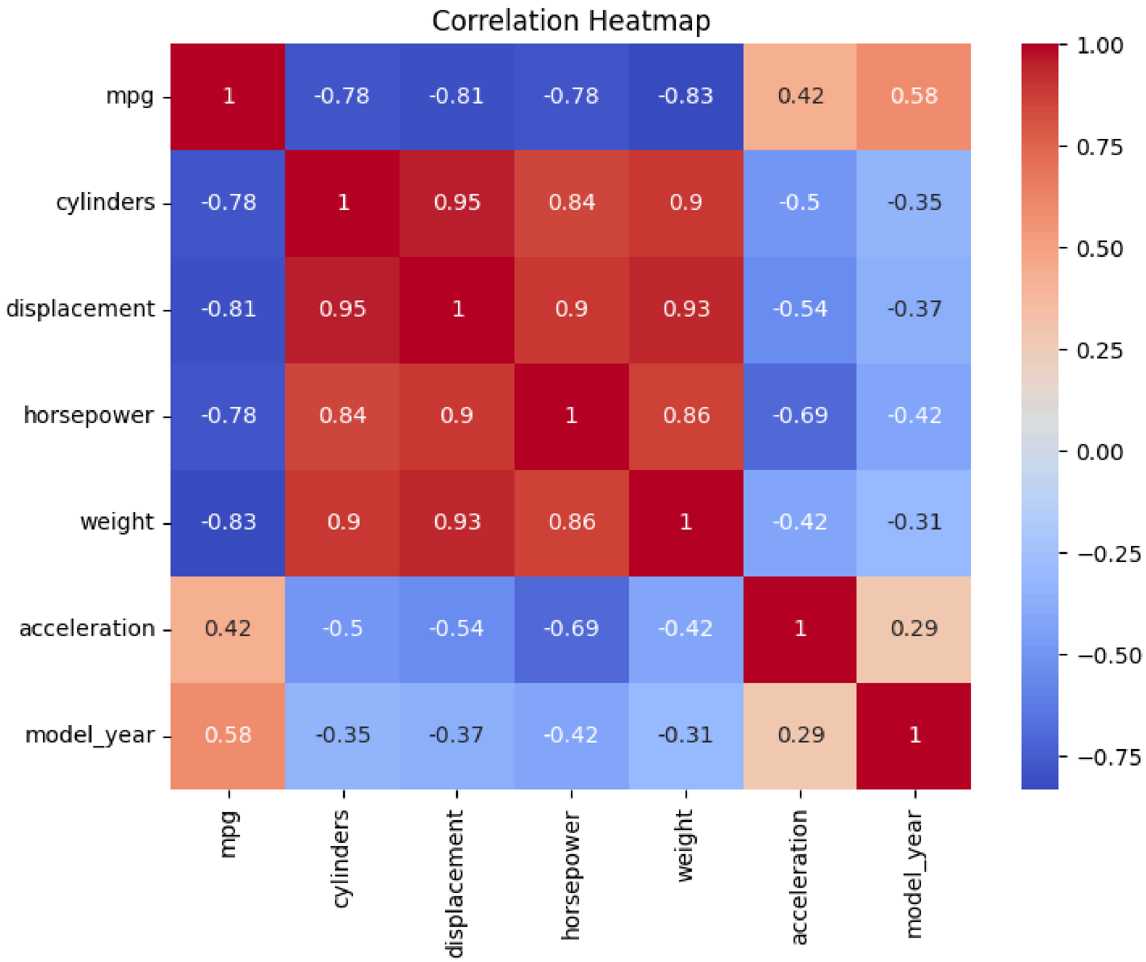
```
df.hist(figsize=(12,8), bins=20)
plt.suptitle("Histograms of Numerical Features", fontsize=15)
plt.show()
```



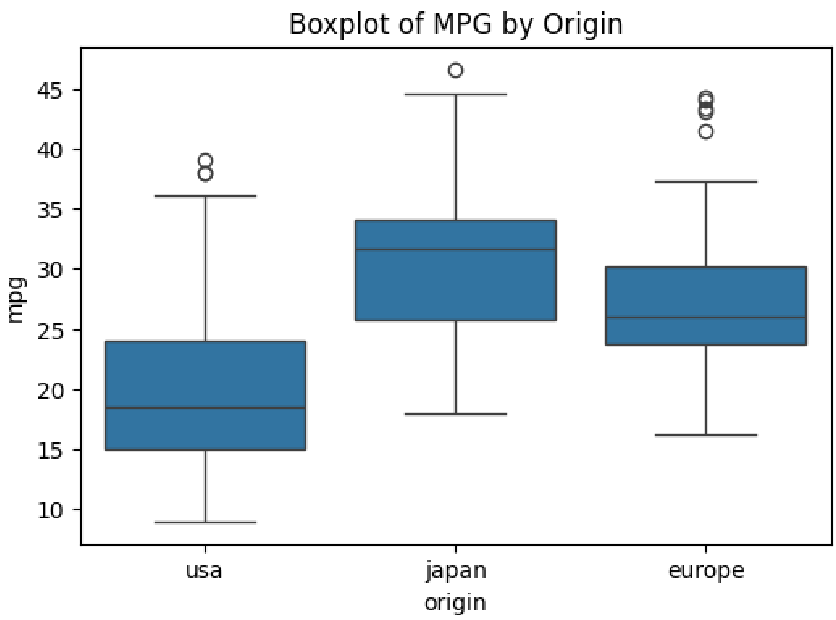

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='horsepower', y='mpg', data=df)
plt.title("Horsepower vs MPG")
plt.show()
```



```
plt.figure(figsize=(8,6))
numerical_df = df.drop(['origin', 'name'], axis=1)
sns.heatmap(numerical_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.boxplot(x='origin', y='mpg', data=df)
plt.title("Boxplot of MPG by Origin")
plt.show()
```




```
X = df[['horsepower', 'weight', 'acceleration', 'displacement']]
y = df['mpg']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

```
print("\nModel Evaluation:")
print("Mean Squared Error (MSE):", mse)
print("R² Score:", r2)
```

```
Model Evaluation:
Mean Squared Error (MSE): 18.065921762583905
R² Score: 0.6460478972647503
```

```
plt.figure(figsize=(6,4))
sns.scatterplot(x=y_test, y=y_pred)
plt.xlabel("Actual MPG")
plt.ylabel("Predicted MPG")
plt.title("Actual vs Predicted MPG")
plt.show()
```